BUSINESS ANALYSIS DOCUMENT PACK

Automated Payment Reconciliation Engine

FinTech / Payments | Project code PRE-2026

Complete set of Business Analyst deliverables for one FinTech project: from the first stakeholder interview to the benefits report after go-live. Every document uses the same worked example, so one requirement can be traced from a business pain to a tested, signed-off feature.

Client problem	Payment mismatches identified 3 days late; 4 analysts spend 6 hours daily on manual reconciliation
Solution	Automated reconciliation engine with exception queue, audit log and daily report
Result	24 hours per day reduced to 4 hours (83% reduction); mismatches visible on the same day
Documents	17 documents including BRD, FRD, process flows, sequence diagrams, RTM, test cases and UAT

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Document Control

Project	Automated Payment Reconciliation Engine (PRE-2026)
Domain	FinTech / Payments (UPI, Cards, Net Banking)
Sponsor	Finance Controller
Product Owner	Head of Payments Product
Prepared by	Business Analyst
Pack version	1.0
Status	Requirements baselined; UAT approved 15-Oct; benefits tracking in progress
Distribution	Sponsor, Product Owner, Finance, Operations, Tech Team, QA, Compliance / Audit

Revision history

Version	Stage	Description of change	Author
0.1	Draft	First draft after discovery interviews and data profiling	Business Analyst
0.5	Reviewed	Stakeholder review comments incorporated; duplicate rule boundary clarified (10 minutes inclusive)	Business Analyst
0.9	UAT	Updated after system test: defect DEF-007 root cause recorded; UAT scenarios added	Business Analyst
1.0	Baselined	Approved scope and requirements; UAT sign-off recorded (15-Oct)	Business Analyst

Approvals

Name	Role	Decision	Signature	Date
	Finance Controller (Sponsor)	Approved		
	Product Owner	Approved		
	Tech Lead	Reviewed		
	Compliance / Audit	Reviewed		

How to read this pack

The pack follows the life cycle of one project, so a single requirement can be traced from the business problem to the tested and signed-off feature. Documents are numbered in the order a Business Analyst produces them. Requirement identifiers are used consistently: OBJ (business objective), BR (business requirement), RULE (business rule), UC (use case), FR (functional requirement), NFR (non-functional requirement), US (user story), TC (test case), UAT (acceptance test).

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1	Problem and Opportunity Brief	PRE-DOC-01	Sponsor, Product Owner
2	Elicitation Plan and Interview Notes	PRE-DOC-02	BA, Finance
3	As-Is Analysis, Root Cause and Gap Analysis	PRE-DOC-03	All stakeholders
4	Stakeholder Analysis and RACI	PRE-DOC-04	Sponsor, PMO
5	Scope Statement	PRE-DOC-05	All stakeholders
6	Business Requirements Document (BRD)	BRD-PRE-001	Sponsor, Finance
7	Functional Requirements Document (FRD / SRS)	FRD-PRE-001	Developers, QA
8	Process Flows (As-Is and To-Be)	PRE-DOC-08	All stakeholders
9	Sequence Diagrams	PRE-DOC-09	Developers, QA
10	User Stories and Acceptance Criteria	PRE-DOC-10	Scrum team
11	Requirements Traceability Matrix (RTM)	RTM-PRE-001	QA, Audit
12	Test Plan, Test Cases and Defect Report	TST-PRE-001	QA, BA
13	User Acceptance Testing (UAT) Document	UAT-PRE-001	Finance, Sponsor
14	Go-Live Checklist, Release Notes and Hypercare Plan	PRE-DOC-14	Operations, Tech
15	Benefits Realization Report	PRE-DOC-15	Sponsor, Leadership
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1. Problem and Opportunity Brief

Document ID	PRE-DOC-01
Document title	Problem and Opportunity Brief
Version / Status	1.0 / Approved
Author / Owner	Business Analyst
Reviewers	Finance Controller, Product Owner, Tech Lead

1.1 Purpose

This brief explains why the project exists. It states the business problem, the cost of leaving it unsolved, the opportunity, and the objectives the project must achieve. It is the document the sponsor uses to approve the start of requirements work.

1.2 Business context

The company accepts UPI, card and net banking payments through a payment gateway (for example Razorpay) and receives settlement from partner banks. Every working day the Finance team must prove that the amounts the gateway reports as successful agree with the amounts the bank has credited. This proof is called payment reconciliation. Gaps delay merchant payouts, create audit findings and tie up skilled analysts.

1.3 Problem statement

Problem: Payment mismatches are identified about 3 days after the transaction because reconciliation is done manually in Excel. The process is slow, error-prone and delays settlement.

1.4 Worked example of the problem

Source	Amount	Comment
Razorpay (gateway) report	₹10,00,000	Total of transactions reported as successful
Bank settlement credit	₹9,95,000	Amount actually credited
Difference	**₹5,000**	Must be explained before settlement can be confirmed

The analyst must find the cause of the ₹5,000 gap. Possible causes are a duplicate transaction, a missing settlement, a wrong status, a refund, or a fee deduction. Today the search is done by hand, file by file, followed by email to the gateway team, the bank and the merchant.

1.5 Current-state impact

Measure	Current value	Business impact
Analysts on reconciliation	4	Skilled staff on repetitive work
Hours per analyst per day	6	Most of the working day
Total effort	24 hours per day	High operating cost
Time to detect a mismatch	About 3 days	Late settlement, merchant escalations
Error pattern	Missed duplicates, wrong matches	Audit and financial exposure

1.6 Opportunity

Automate matching so that the system handles every clean transaction, detects duplicates, missing settlements and status mismatches on the same day, and sends only exceptions to analysts. Analysts then spend their time on investigation and merchant support instead of file matching.

1.7 Business objectives (SMART)

ID	Objective	Measure	Baseline	Target
OBJ-1	Reduce daily reconciliation effort	Analyst hours per day	24	4 or less
OBJ-2	Detect mismatches sooner	Days from transaction to detection	About 3	Same day (T+0)
OBJ-3	Reduce manual errors	Missed or wrong matches	Audit baseline	90% fewer

ID	Objective	Measure	Baseline	Target
OBJ-4	Speed up merchant settlement	Settlements delayed by open mismatches	Delay report baseline	50% fewer

1.8 Assumptions and constraints

- Gateway and bank files arrive daily by 10 am in an agreed format.
- The gateway transaction ID can be linked to the bank settlement reference.
- A gateway-to-bank difference can include legitimate items (MDR fee, GST on fee, refunds). The system must classify these rather than treat every difference as an error. *To be confirmed with Finance in the first workshop.*
- Go-live must not disturb month-end close.

1.9 High-level risks

Risk	Impact	Mitigation
Gateway file format changes without notice	Matching breaks	Versioned file contract; schema validation; alert on failure
Poor data quality in bank files	False exceptions	Data profiling in discovery; tolerance rules
Analysts distrust automated matches	Low adoption	Parallel run; visible match logic; early UAT involvement

1.10 Recommendation

Proceed to requirements elicitation. Expected outcome: effort reduced from 24 to about 4 hours per day, mismatches visible on the transaction day, and an audit trail for every decision.

2. Elicitation Plan and Interview Notes

Document ID	PRE-DOC-02
Document title	Elicitation Plan and Interview Notes
Version / Status	1.0 / Approved
Author / Owner	Business Analyst
Reviewers	Finance Controller, Product Owner, Tech Lead

2.1 Objective

To understand the real reconciliation process, its pain points and its data, and to collect requirements from every group that uses or is affected by the process.

2.2 Elicitation plan

Technique	Participants	Goal	Timing	Output
One-to-one interview	Finance Manager	End-to-end process, pain, success measures	Week 1	Interview notes
Group workshop	Operations team	Daily steps and exception types	Week 1	Workshop notes, issue list
Interview	Payment Gateway (PSP) team	File formats, status codes, delivery time	Week 2	File specification
Observation (shadowing)	Two analysts	See the real process, not the described one	Week 2	Observed process steps
Document analysis	Sample files, past mismatch emails	Find patterns and volumes	Week 2	Pattern summary
Data analysis (SQL / Excel)	BA with 3 months of files	Quantify mismatch causes	Weeks 2-3	Data profile report
Requirements workshop	All key stakeholders	Agree and prioritise requirements	Week 3	Prioritised requirement list

2.3 Interview guide

Finance Manager

- How does reconciliation work today, step by step?
- What would a successful day look like for you?
- What causes most mismatches?
- Which reports do you use and who owns them?
- Which decisions are delayed because mismatches are found late?
- What are the month-end and audit expectations?

Operations team

- How do you start your day? Which files do you open first?
- Show me how you match one transaction.
- What do you do when amounts do not match?
- Which cases take the longest to resolve?
- Which emails or chasers do you send most often?

Payment Gateway (PSP) team

- Which file formats and delivery times do you support?
- Which status codes exist (success, failed, pending, refunded)?
- How are refunds and chargebacks reported?
- Can you provide an API as well as files?

2.4 Interview and workshop log

Date	Interviewee	Key insight	Follow-up action
Week 1	Finance Manager	Mismatches are found on day 3 because the gateway file is compared only after the bank credit arrives	Obtain bank credit schedule
Week 1	Operations Analyst	About 30 percent of time goes on duplicates and status errors	Pull duplicate sample from last 3 months
Week 1	Operations team (workshop)	Exceptions are chased by email and tracked in personal sheets; no single owner	Define exception queue and status model
Week 2	PSP team	Settlement file arrives by 10 am; refunds come in a separate file	Request file specification and test files
Week 2	Analyst (observation)	Matching uses VLOOKUP on transaction ID; formula breaks when rows are filtered	Record as evidence for error root cause

2.5 Requirements workshop agenda (Week 3)

Time	Topic	Lead	Outcome
0:00 - 0:10	Objectives and agenda	BA	Shared understanding
0:10 - 0:30	Review of root causes and data profile	BA	Agreement on problems
0:30 - 1:00	Walk through draft business rules (duplicate, match, missing, status)	BA / Finance Manager	Confirmed rules and boundaries
1:00 - 1:30	Scope: in and out	Product Owner	Scope agreed
1:30 - 1:50	Prioritisation (MoSCoW)	All	Ranked requirements
1:50 - 2:00	Actions, next steps	BA	Action list

3. As-Is Analysis, Root Cause and Gap Analysis

Document ID	PRE-DOC-03
Document title	As-Is Analysis, Root Cause and Gap Analysis
Version / Status	1.0 / Approved
Author / Owner	Business Analyst
Reviewers	Finance Controller, Product Owner, Tech Lead

3.1 Current-state process

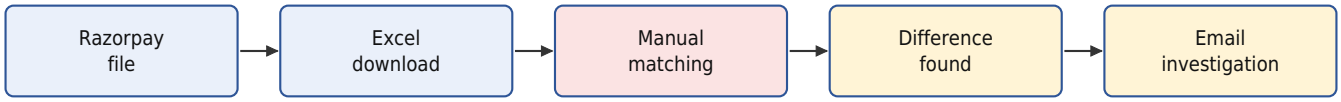


Figure 3.1: As-Is reconciliation flow

Step	Activity	Performed by	Tools	Pain point
1	Download gateway transaction file	Analyst	Gateway portal	Manual, repeated daily
2	Open in Excel and combine with bank file	Analyst	Excel	Large files, slow
3	Match by transaction ID and amount (VLOOKUP, filters)	Analyst	Excel	Formula errors, missed rows
4	Identify differences	Analyst	Excel	Found only after bank credit (T+3)
5	Email PSP, bank or merchant to investigate	Analyst	Email	No owner, no status, no audit trail

3.2 Problems discovered

No.	Problem	Evidence	Effect
P1	Duplicate transactions	Same merchant, amount and order ID appear twice in the gateway file	Overstated gateway total and false differences
P2	Missing settlements	Gateway success with no bank credit	Delayed merchant payout
P3	Wrong transaction status	Gateway success but bank rejected, or the reverse	Incorrect ledger entries
P4	Human errors	Manual lookups, copy and paste, broken formulas	Wrong or missed matches
P5	Late detection	Files compared only after bank credit	Issues known on day 3

3.3 Root cause analysis (5 Whys on the ₹5,000 difference)

Why	Answer
1. Why is there a ₹5,000 difference?	A gateway transaction has no matching bank credit.
2. Why is there no matching credit?	It was a duplicate; the bank settled the payment once.
3. Why was a duplicate created?	The customer retried and the gateway recorded two successes.
4. Why was it not caught?	Matching is manual and done only after the bank credit arrives.
5. Why is matching manual?	There is no automated, rule-based reconciliation or exception queue.

Root cause: the absence of automated, rule-based reconciliation with an exception queue.

3.4 Gap analysis

Area	Current state	Required state	Gap / action
Matching	Manual in Excel	Automated engine	Build matching engine
Duplicate detection	Found by eye	Rule-based auto-flag	Define and implement RULE-01
Missing settlement	Found late	Detected on the same day	Implement RULE-03 with cut-off time
Exception handling	Email chains	Queue with owner and status	Build queue and workflow
Timing	T+3	T+0	Scheduled daily run
Audit trail	Scattered emails	Central log of every decision	Add audit log
Reporting	None	Daily summary	Build summary report

3.5 Data profile summary (discovery)

Three months of gateway and bank files were profiled with SQL to size the problem before requirements were fixed.

```
-- Potential duplicates in historical gateway data
SELECT merchant_id, order_id, amount, COUNT(*) AS cnt
FROM gateway_txn
GROUP BY merchant_id, order_id, amount
HAVING COUNT(*) > 1;

-- Gateway successes with no bank credit
SELECT g.transaction_id, g.amount
FROM gateway_txn g
LEFT JOIN bank_settlement b ON b.settlement_ref = g.transaction_id
WHERE g.gateway_status = 'SUCCESS' AND b.settlement_ref IS NULL;
```

4. Stakeholder Analysis and RACI

Document ID	PRE-DOC-04
Document title	Stakeholder Analysis and RACI
Version / Status	1.0 / Approved
Author / Owner	Business Analyst
Reviewers	Finance Controller, Product Owner, Tech Lead

4.1 Stakeholder register

Stakeholder	Role in project	Why they care	Power	Interest	Engagement approach
Finance Controller	Sponsor	Accurate settlement, clean audits	High	High	Manage closely: weekly review
Product Owner	Approver	Owns scope and priority	High	High	Manage closely: twice-weekly meeting
Operations Analysts	Daily users	Their daily work changes	Low	High	Keep informed; involve in UAT
Support Team	Handles queries	Fewer escalations	Low	High	Keep informed; demos
PSP (Gateway) Team	Data provider	Sends the transaction file	Medium	Medium	Consult on file contract
Tech Team	Builders	Needs unambiguous requirements	Medium	High	Collaborate daily
Compliance / Audit	Reviewer	Evidence and controls	High	Low	Keep satisfied: milestone reviews

4.2 Power-interest grid

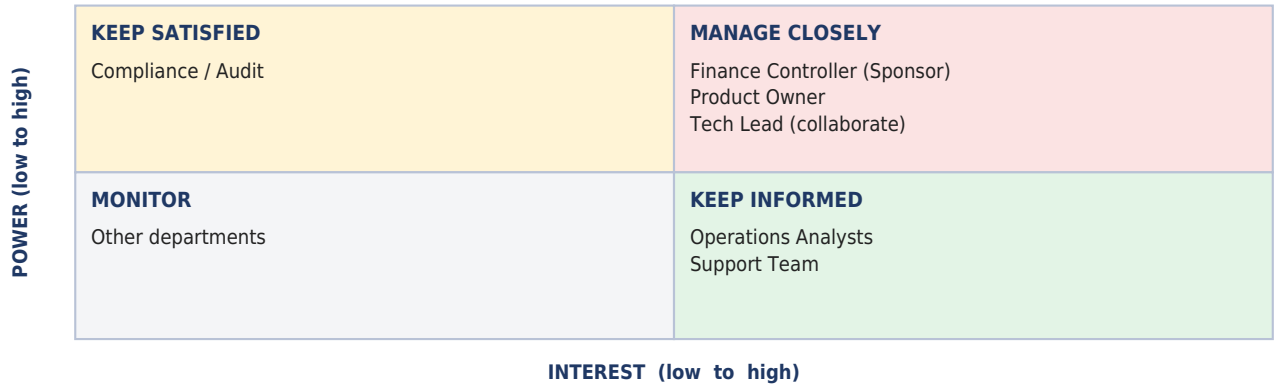


Figure 4.1: Stakeholder power-interest grid

4.3 RACI matrix

R = Responsible, A = Accountable, C = Consulted, I = Informed.

Activity	BA	Product Owner	Finance Controller	Tech Lead	QA	PSP Team	Analysts
Elicit requirements	R	A	C	C	I	C	C
Approve scope	R	A	C	I	I	I	I
Write BRD / FRD	R	A	C	C	C	I	I
Design and build	C	I	I	A/R	C	C	I
Write test cases	C	I	I	C	A/R	I	I
Execute UAT	C	C	A	I	R	I	R
UAT sign-off	R (collects)	C	A	I	I	I	C
Go-live	R	A	C	R	R	C	I
Measure benefits	R	C	A	I	I	I	C

5. Scope Statement

Document ID	PRE-DOC-05
Document title	Scope Statement
Version / Status	1.0 / Approved
Author / Owner	Business Analyst
Reviewers	Finance Controller, Product Owner, Tech Lead

5.1 Project scope

The project delivers an automated reconciliation engine for gateway-to-bank payment reconciliation, including exception management, an audit log and reporting.

5.2 In scope

Item	Detail
UPI	Gateway-to-bank reconciliation
Cards	Credit and debit card transactions
Net Banking	Bank-redirect payments
Duplicate detection	Rule-based, automatic (RULE-01)
Missing settlement detection	Gateway success with no bank credit (RULE-03)
Status mismatch detection	Gateway status compared with bank status (RULE-04)
Exception queue	Review, assignment, comments, resolution notes
Daily reconciliation report	Matched, unmatched, exceptions, value
Audit log	Every automated and manual decision
Alerts	Late or invalid file notifications

5.3 Out of scope

Item	Reason
Wallet payments	Different settlement flow; separate phase
Mobile app changes	No user-facing change needed
Merchant portal changes	Merchants see no change in this release
Chargeback management	Planned for a future release
Accounting ledger posting	Handled by Finance systems

5.4 Deliverables

- Approved BRD and FRD with traceability matrix
- Reconciliation engine, exception queue screens and reports
- Test evidence and UAT sign-off
- User guide, training and hypercare support
- Benefits realization report

5.5 Acceptance criteria for the project

- All Must requirements delivered and accepted in UAT
- No open critical or high defects at go-live
- Daily reconciliation effort at or below 4 hours
- Mismatches visible on the transaction day

5.6 Scope change control

After baseline, any addition or change requires a Change Request describing the change, the reason, the impact on time, cost and risk, and written approval from the Product Owner. Approved changes update the BRD, FRD and RTM.

6. Business Requirements Document (BRD)

Document ID	BRD-PRE-001
Document title	Business Requirements Document
Version / Status	1.0 / Baselined
Author / Owner	Business Analyst
Reviewers	Finance Controller, Product Owner, Tech Lead, Compliance

6.1 Purpose

This BRD states what the business needs from the project. It describes business objectives, scope, requirements and rules. It does not describe how the system is built; that is covered in the FRD (document 7).

6.2 Executive summary

Finance reconciles payment gateway files with bank settlement files manually in Excel. Four analysts spend six hours each per day (24 hours in total), mismatches are found about three days late, and errors are frequent. The project introduces an Automated Payment Reconciliation Engine that matches transactions daily, flags duplicates, missing settlements and status mismatches, and passes only exceptions to analysts. The expected result is four hours of daily effort (an 83 percent reduction) and same-day detection of mismatches.

6.3 Background and business context

The company processes UPI, card and net banking payments through a payment gateway and receives settlement from partner banks. Daily reconciliation proves that what the gateway reports as successful equals what the bank credited. Gaps delay merchant payouts and create audit findings.

6.4 Problem statement

Manual, Excel-based reconciliation is slow, error-prone and late. For example, a gateway total of ₹10,00,000 against a bank credit of ₹9,95,000 leaves a ₹5,000 difference that can take hours to explain.

6.5 Business objectives and success measures

ID	Objective	Baseline	Target	How measured
OBJ-1	Reduce daily reconciliation effort	24 hours per day	4 hours or less	Timesheet sample and engine usage log
OBJ-2	Detect mismatches earlier	About 3 days	Same day (T+0)	Time from transaction to exception creation
OBJ-3	Reduce manual errors	Baseline from 3-month audit	90% fewer	Post go-live audit sample
OBJ-4	Faster settlement to merchants	Baseline from delay report	50% fewer delayed settlements	Settlement delay report

6.6 Scope

In scope	Out of scope
UPI, cards, net banking reconciliation Duplicate, missing settlement and status mismatch detection Exception queue and workflow Audit log Daily reconciliation report and alerts	Wallet payments Mobile app changes Merchant portal changes Chargeback management Accounting ledger posting

6.7 Stakeholders

Stakeholder	Interest	Influence
Finance Controller (Sponsor)	Accurate settlement and audit readiness	High
Product Owner	Scope, priorities, approval	High
Operations Analysts	Daily users of the solution	Low (high interest)
Support Team	Fewer merchant escalations	Low (high interest)
PSP Team	Provides the transaction file	Medium
Tech Team	Builds the solution	Medium

Stakeholder	Interest	Influence
Compliance / Audit	Controls and evidence	High (low interest)

6.8 Current state and future state

Aspect	Current state (As-Is)	Future state (To-Be)
Matching	Manual in Excel	Automated reconciliation engine
Duplicates	Found by eye	Auto-flagged by rule
Exceptions	Email chains	Queue with owner and status
Detection time	About T+3	T+0
Evidence	Scattered emails	Central audit log
Effort	24 hours per day	About 4 hours per day

6.9 Business requirements

ID	Business requirement	Objective	Priority	Source	Acceptance measure
BR-01	The system shall reduce the effort needed for daily payment reconciliation	OBJ-1	Must	Finance Manager	Daily effort 4 hours or less
BR-02	The business shall identify mismatches on the same day as the transaction	OBJ-2	Must	Finance Controller	Exceptions created by end of T+0
BR-03	The business shall reduce incorrect or missed matches caused by manual work	OBJ-3	Must	Finance Controller	90% error reduction
BR-04	The business shall review and resolve exceptions with a clear owner and status	OBJ-4	Must	Operations team	Every exception has an owner and a note on closure
BR-05	The business shall keep an auditable record of every reconciliation decision	Compliance	Should	Compliance	Audit log complete and read only
BR-06	The business shall see a daily summary of reconciliation health	OBJ-2	Should	Finance Controller	Report available by 11 am daily

6.10 Business rules

ID	Business rule	Example
RULE-01	Duplicate = same Merchant ID + same Amount + same Order ID within 10 minutes (inclusive)	M123, O111, ₹1,000 at 10:00 and 10:06 = Duplicate
RULE-02	Matched = gateway transaction found in the bank settlement and amounts agree within the approved tolerance	TXN0001 ₹1,000 in both files = Matched
RULE-03	Missing settlement = gateway status SUCCESS and no bank credit by the cut-off time	SUCCESS at gateway, no bank record by cut-off
RULE-04	Status mismatch = gateway status differs from bank status for the same transaction	Gateway SUCCESS, bank REJECTED
RULE-05	An exception cannot be closed without a resolution note	Resolve with empty note is blocked

6.11 Assumptions, constraints and dependencies

Type	Item
Assumption	Gateway and bank files arrive daily in the agreed format
Assumption	The transaction ID links gateway and bank records
Constraint	No go-live during month-end close
Constraint	Personal and card data must follow data protection rules
Dependency	PSP provides the file specification and test files
Dependency	Bank provides the settlement file by 10 am

6.12 Risks

ID	Risk	Likelihood	Impact	Mitigation	Owner
R1	Gateway file format changes without notice	Medium	High	Versioned contract; schema validation	Tech Lead
R2	Poor bank data quality	Medium	Medium	Data profiling; tolerance rules	BA
R3	Low user adoption	Medium	High	Parallel run; early UAT	Product Owner

ID	Risk	Likelihood	Impact	Mitigation	Owner
R4	Late files	High	Medium	Late-file alert; manual fallback	Operations Lead

6.13 Benefits and cost-benefit view

Item	Value
Hours saved per day	20 (24 minus 4)
Hours saved per year	About 5,000 (assuming 250 working days)
Cost per hour	Finance to supply the loaded rate
Non-financial benefits	Faster settlement, fewer audit issues, better merchant trust

6.14 High-level timeline

Phase	Activity	Duration
1	Discovery and elicitation	3 weeks
2	Requirements and design	3 weeks
3	Build (sprints)	8 weeks
4	System test	2 weeks
5	UAT and parallel run	2 weeks
6	Go-live and hypercare	2 weeks

6.15 Approval

Name	Role	Decision	Signature	Date
	Finance Controller	Approved		
	Product Owner	Approved		
	Tech Lead	Reviewed		

7. Functional Requirements Document (FRD / SRS)

Document ID	FRD-PRE-001
Document title	Functional Requirements Document
Version / Status	1.0 / Baselined
Author / Owner	Business Analyst
Reviewers	Product Owner, Tech Lead, QA Lead
Derived from	BRD-PRE-001

7.1 Introduction

This document describes what the system shall do. It is written for developers, testers and the BA. Every functional requirement traces to a business requirement in the BRD.

7.2 System overview

The Reconciliation Engine reads the gateway transaction file and the bank settlement file, applies matching rules, classifies every transaction, and presents exceptions to analysts through a web screen. A scheduler runs the process every day.

7.3 Actors

Actor	Description
Scheduler	Starts the daily ingestion and reconciliation run
Finance Analyst	Reviews and resolves exceptions
Operations Lead	Receives alerts, assigns work, can start a manual run
Finance Controller	Views the daily summary
Auditor	Reads the audit log
Gateway / Bank	External providers of the daily files

7.4 Use case list

ID	Use case	Primary actor	Traces to
UC-01	Ingest and validate files	Scheduler	FR-01, FR-02, FR-11
UC-02	Run reconciliation	Scheduler	FR-03 to FR-07, FR-09
UC-03	Review exception	Finance Analyst	FR-07
UC-04	Resolve exception	Finance Analyst	FR-08, FR-09
UC-05	View daily summary	Finance Controller	FR-10
UC-06	Review audit log	Auditor	FR-09

7.5 Use case specifications

UC-01 Ingest and validate files

Trigger	Scheduler at 10:30 daily, or files arriving on the SFTP location
Precondition	Gateway and bank have delivered files
Main flow	1. System picks up the gateway file and the bank file. 2. System validates file name, columns, data types and row count. 3. System stores the records. 4. System writes an audit entry for each file.
Alternate flow	File missing or invalid: system stops the run for that file and sends an alert to the Operations Lead (FR-11).
Postcondition	Valid records are available for reconciliation

UC-02 Run reconciliation

Trigger	Scheduler at 11:00 daily, or manual run by the Operations Lead
Precondition	Gateway and bank files are loaded and validated

Main flow	1. Engine selects the day's gateway transactions. 2. Engine checks each for duplicates (RULE-01). 3. Engine matches the remaining transactions to bank records (RULE-02). 4. Engine checks for missing settlement (RULE-03) and status mismatch (RULE-04). 5. Engine marks matched items MATCHED and closes them automatically. 6. Engine places every other item in the Exception Queue. 7. Engine writes audit entries and the summary report.
Alternate flow A	Files not valid: run does not start; alert sent to Operations Lead.
Alternate flow B	Amount differs within the approved tolerance: classified MATCHED with a tolerance note.
Exceptions	Database unavailable: run marked Failed and alert sent.
Postcondition	Every transaction has a recon_status; summary report is available

UC-04 Resolve exception

Trigger	Analyst opens an exception from the queue
Precondition	Analyst has the Analyst role; exception status is Open or In Progress
Main flow	1. Analyst selects the exception. 2. System shows the gateway and bank records side by side. 3. Analyst assigns it to self. 4. Analyst selects a resolution reason and enters a note. 5. Analyst clicks Resolve. 6. System sets status Resolved and writes an audit entry.
Alternate flow	Note is empty: system blocks closure and shows a message (RULE-05).
Postcondition	Exception is Resolved with note, user and time recorded

7.6 Functional requirements

ID	Requirement	Input	Processing	Output / error handling	Traces to	Priority
FR-01	The system shall ingest the gateway transaction file daily and validate its format	CSV or JSON file	Validate columns, types, row count	Records stored; invalid file rejected and alert sent	BR-01, BR-02	Must
FR-02	The system shall ingest the bank settlement file daily and validate its format	Settlement file	Validate and parse	Records stored; invalid file rejected and alert sent	BR-01, BR-02	Must
FR-03	The system shall match each gateway transaction to a bank record per RULE-02	Gateway and bank records	Compare transaction ID and amount	recon_status = MATCHED; unmatched go to queue	BR-01, BR-03	Must
FR-04	The system shall detect duplicate transactions per RULE-01 and flag them as Duplicate	Gateway records	Apply RULE-01	recon_status = DUPLICATE with reference to the original	BR-03	Must
FR-05	The system shall detect missing settlements per RULE-03	SUCCESS records without bank credit	Apply RULE-03 after cut-off	recon_status = MISSING_SETTLEMENT	BR-02	Must
FR-06	The system shall detect status mismatches per RULE-04	Gateway and bank status	Apply RULE-04	recon_status = STATUS_MISMATCH	BR-03	Must
FR-07	The system shall place every unmatched or flagged item in the Exception Queue	Flagged records	Create a queue item, status Open	Queue entry	BR-04	Must
FR-08	The system shall allow an analyst to assign, comment on and resolve an exception	Analyst action	Update owner, status, note	Updated record; block if note empty	BR-04	Must
FR-09	The system shall record every automated and manual action in an audit log	Any action	Write who, what, when, old and new value	Read-only log; retry and alert on failure	BR-05	Should
FR-10	The system shall produce a daily reconciliation summary report	Day's results	Aggregate counts and values	Report on screen and download	BR-06	Should
FR-11	The system shall alert the Operations Lead when a file is late, missing or invalid	File status	Check at 10:30 and 11:00	Email alert; retry	BR-02	Should

7.7 Duplicate detection logic (RULE-01)

```
For each incoming transaction T:
    Find an earlier transaction E where
        E.merchant_id = T.merchant_id
    AND E.order_id    = T.order_id
    AND E.amount      = T.amount
    AND T.timestamp - E.timestamp <= 10 minutes
    If found:
        Mark T as DUPLICATE (duplicate_of = E)
        Send T to the Exception Queue
    Else:
        Continue normal matching
```

Edge case	Decision (confirmed with Finance)
Same order and amount exactly 10 minutes apart	Counts as duplicate (window is inclusive)
Same order and amount after 10 minutes	Not a duplicate; process normally
Same order, different amount	Not a duplicate; may be a partial payment
Different merchant, same order ID	Not a duplicate

7.8 Data dictionary

Field	Type	Source	Description	Rules
merchant_id	String(10)	Gateway	Unique merchant identifier	Mandatory, e.g. M123
order_id	String(20)	Gateway	Merchant order reference	Mandatory, e.g. O111
transaction_id	String(24)	Gateway	Gateway transaction reference	Mandatory, unique per attempt
amount	Decimal(12,2)	Gateway / Bank	Amount in INR	Greater than 0
payment_mode	Enum	Gateway	UPI, CARD, NETBANKING	Mandatory
gateway_status	Enum	Gateway	SUCCESS, FAILED, PENDING, REFUNDED	Mandatory
bank_status	Enum	Bank	CREDITED, REJECTED, NOT_FOUND	Derived from bank file
txn_timestamp	Datetime	Gateway	When the transaction occurred	IST, ISO 8601
settlement_ref	String(30)	Bank	Bank settlement reference	Links to transaction_id
recon_status	Enum	System	MATCHED, DUPLICATE, MISSING_SETTLEMENT, STATUS_MISMATCH	Set by engine
exception_owner	String	System	Analyst assigned	Null until assigned
resolution_note	Text	Analyst	Reason and action taken	Mandatory to close

7.9 API and JSON examples

First transaction:

```
{
  "merchant_id": "M123",
  "order_id": "O111",
  "amount": 1000,
  "transaction_id": "TXN0001",
  "txn_timestamp": "2026-10-12T10:00:00+05:30"
}
```

Second transaction, six minutes later, same merchant, order and amount:

```
{
  "merchant_id": "M123",
  "order_id": "O111",
  "amount": 1000,
  "transaction_id": "TXN0002",
  "txn_timestamp": "2026-10-12T10:06:00+05:30"
}
```

Expected engine output for the second transaction:

```
{
  "transaction_id": "TXN0002",
  "recon_status": "DUPLICATE",
  "duplicate_of": "TXN0001",
  "sent_to_exception_queue": true
}
```

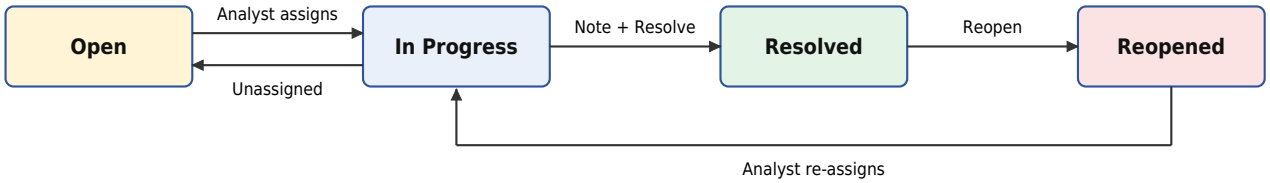
7.10 Screens

Screen	Purpose	Key fields and controls
Exception Queue	List open items	Filters (date, mode, type, owner, status); columns: transaction ID, amount, type, age
Exception Detail	Review one item	Gateway and bank records side by side, history, note box, Assign and Resolve buttons
Daily Summary	Monitor health	Counts and values by status; export button
Audit Log	Evidence	Search by user, date, transaction; read only

7.11 Validation rules

Field	Rule	Message
amount	Numeric, greater than 0, two decimals	Invalid amount
merchant_id	Not empty	Merchant ID required
txn_timestamp	Valid ISO 8601 date and time	Invalid date
resolution_note	Not empty when resolving	A resolution note is required

7.12 Exception status lifecycle



Every flagged item enters the queue as Open.

Figure 7.1: Exception status lifecycle

7.13 Notifications

Event	Recipient	Channel	Content
File late or invalid	Operations Lead	Email	File name, expected time, error
Run failed	Operations Lead, Tech Support	Email	Run ID and error
Exceptions older than 2 days	Finance Manager	Email	Count and list

7.14 Reports

Report	Content	Frequency
Daily Reconciliation Summary	Matched, duplicate, missing, mismatch (count and value); open exceptions	Daily
Exception Ageing	Open exceptions by age bucket	Daily
Audit Extract	Actions in a date range	On demand

7.15 Interfaces

Interface	Direction	Format	Frequency
Gateway (PSP)	Inbound	CSV via SFTP or API	Daily
Bank	Inbound	Settlement file via SFTP	Daily
Email service	Outbound	SMTP	Event based
Single sign-on	Inbound	SAML or OAuth	On login

7.16 Non-functional requirements

ID	Category	Requirement
NFR-01	Performance	Reconcile 1 million transactions in under 30 minutes
NFR-02	Availability	99.5 percent during business hours
NFR-03	Security	Role-based access: Analyst, Operations Lead, Controller, Auditor
NFR-04	Audit	Audit log retained for 8 years and cannot be edited
NFR-05	Data protection	No full card numbers stored; sensitive fields masked
NFR-06	Usability	An analyst can review an exception in 3 clicks or fewer
NFR-07	Scalability	Support 3 times the current daily volume without redesign

8. Process Flows (As-Is and To-Be)

Document ID	PRE-DOC-08
Document title	Process Flows
Version / Status	1.0 / Approved
Author / Owner	Business Analyst
Reviewers	Finance Controller, Product Owner, Tech Lead

8.1 Current state (As-Is)

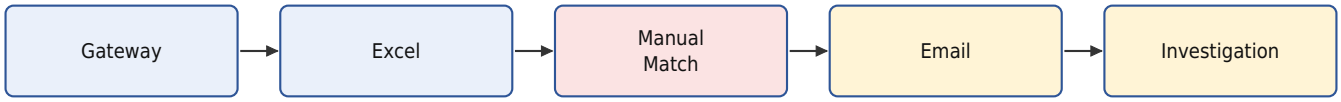


Figure 8.1: As-Is flow

Pain points: manual file handling (Excel step), human error (manual match), slow communication (email step), and no owner or audit trail (investigation).

8.2 Future state (To-Be)

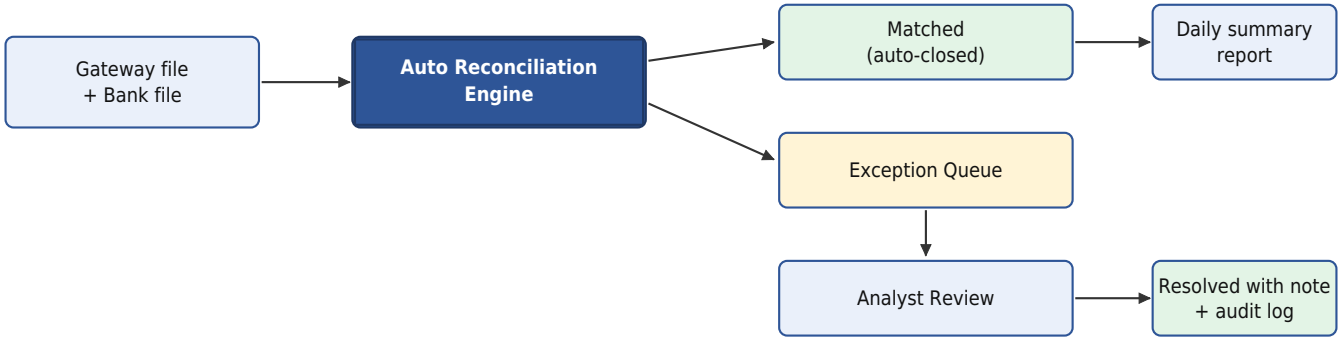


Figure 8.2: To-Be flow with the Auto Reconciliation Engine

8.3 Swimlane summary

Step	Gateway / PSP	Bank	Engine	Analyst	Finance Controller
1	Sends transaction file				
2		Sends settlement file			
3			Validates and loads both files		
4			Applies duplicate, match, missing and status rules		
5			Auto-closes matches; queues exceptions		
6				Reviews and resolves exceptions with note	
7			Produces daily summary		Reviews report

8.4 Impact of the change on people and process

Area	Change	Who is affected	Action
Daily routine	Analysts review exceptions instead of matching every row	Operations Analysts	Training and parallel run
Ownership	Every exception has an owner and status	Operations Lead	Queue assignment procedure
Reporting	Daily summary replaces ad hoc sheets	Finance Controller	Report walkthrough
Audit	Decisions logged centrally	Compliance / Audit	Audit log demonstration

9. Sequence Diagrams

Document ID	PRE-DOC-09
Document title	Sequence Diagrams
Version / Status	1.0 / Approved
Author / Owner	Business Analyst
Reviewers	Finance Controller, Product Owner, Tech Lead
Derived from	FRD-PRE-001

The sequence diagrams show how actors and system components interact over time for the main use cases. Solid arrows are requests; dashed arrows are responses. Shaded bands mark alternative flows.

9.1 Daily file ingestion and reconciliation run (UC-01, UC-02)

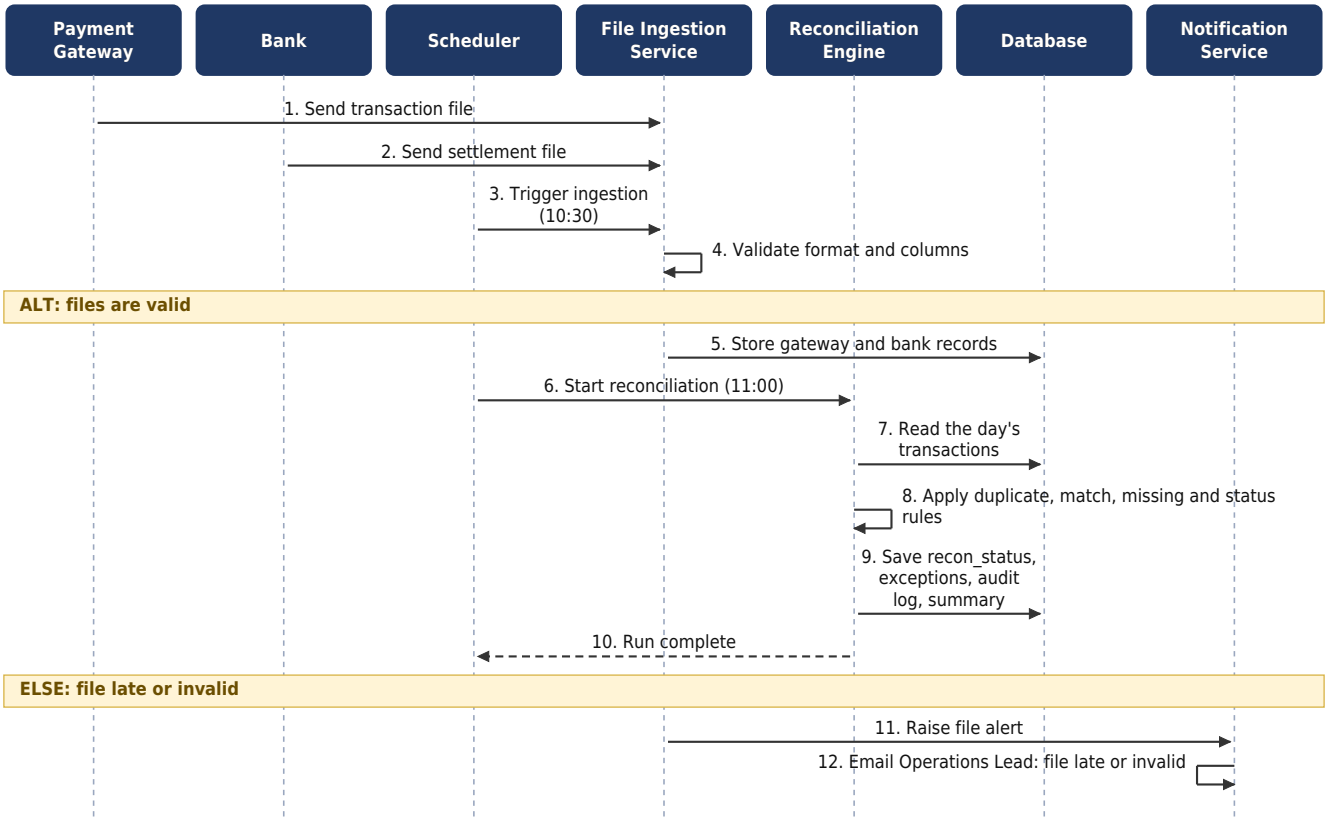


Figure 9.1: Daily ingestion and reconciliation

9.2 Duplicate detection (FR-04, RULE-01)

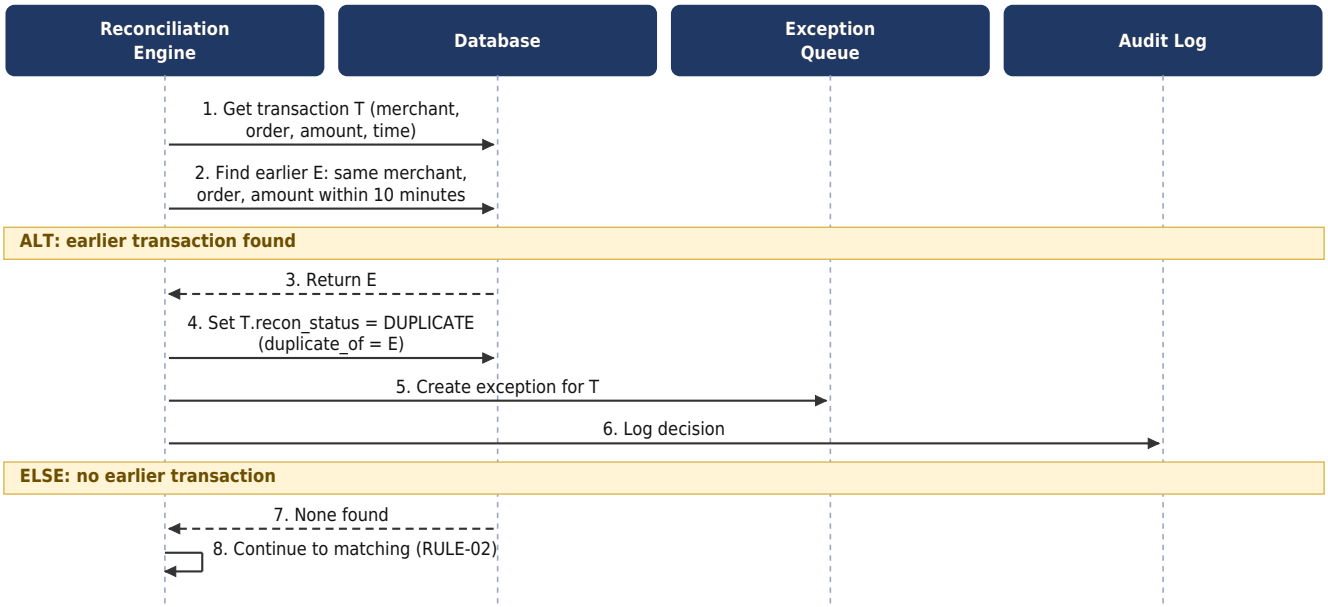


Figure 9.2: Duplicate detection

9.3 Exception review and resolution (UC-03, UC-04)

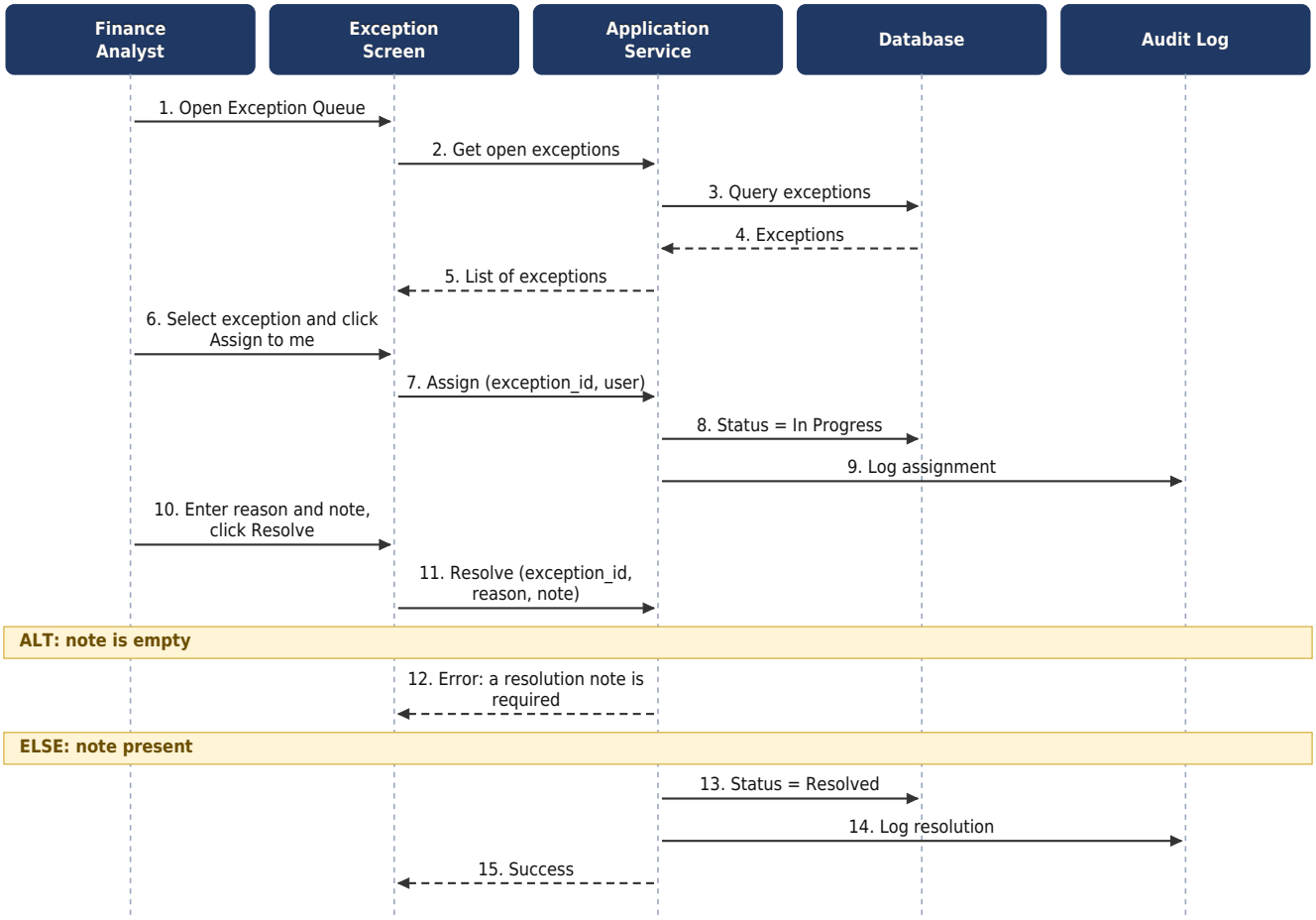


Figure 9.3: Exception review and resolution

9.4 BA and developer clarification (requirement to build)

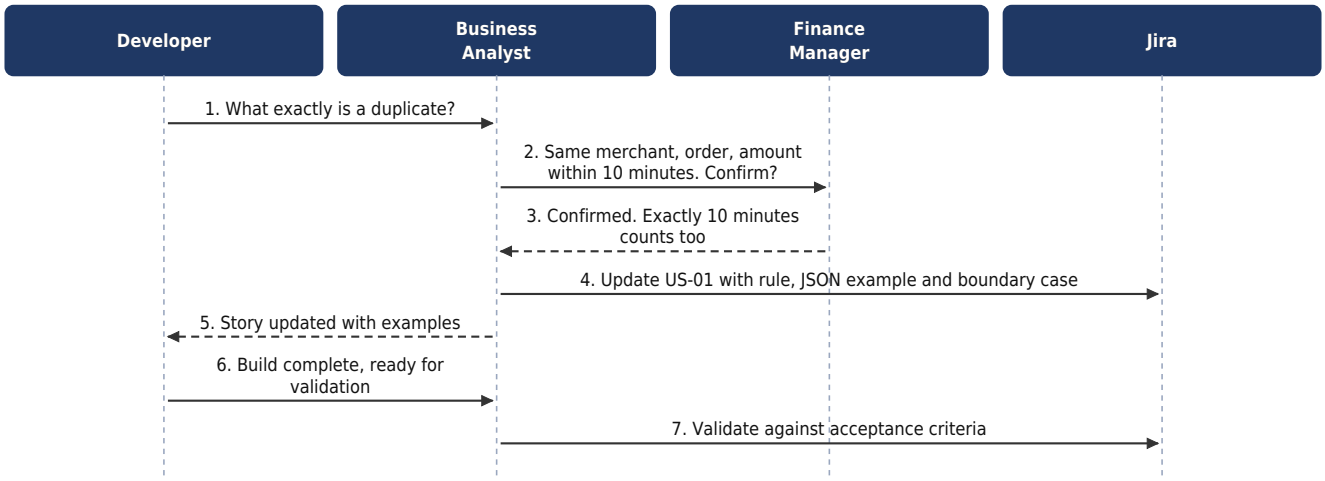


Figure 9.4: BA and developer clarification

9.5 UAT and sign-off

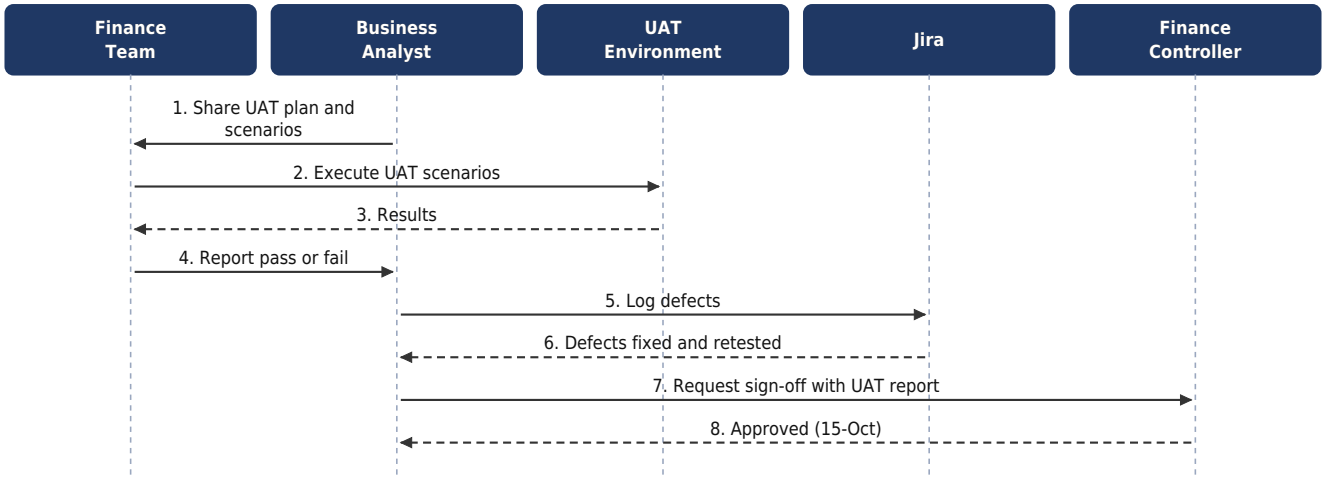


Figure 9.5: UAT and sign-off

10. User Stories and Acceptance Criteria

Document ID	PRE-DOC-10
Document title	User Stories and Acceptance Criteria
Version / Status	1.0 / Approved
Author / Owner	Business Analyst
Reviewers	Finance Controller, Product Owner, Tech Lead
Derived from	FRD-PRE-001

10.1 Epic

EPIC-1 Automated reconciliation: so that analysts spend their time only on exceptions.

10.2 Story backlog

ID	User story	Traces to	Priority	Estimate
US-01	As a Finance Analyst, I want the system to flag duplicate transactions automatically so that I do not need to search for them manually	FR-04, RULE-01	Must	5 points
US-02	As a Finance Analyst, I want unmatched transactions listed in an exception queue so that I can review only what needs attention	FR-07	Must	8 points
US-03	As an Operations Lead, I want an alert when a file is late or invalid so that I can act before the day ends	FR-11	Should	3 points
US-04	As a Finance Controller, I want a daily summary of matched and unmatched value so that I can monitor settlement health	FR-10	Should	5 points
US-05	As an Auditor, I want every decision recorded so that I can verify controls	FR-09	Should	5 points
US-06	As a Finance Analyst, I want to resolve an exception only with a note so that every closure is explained	FR-08, RULE-05	Must	3 points

10.3 Acceptance criteria (Given / When / Then)

US-01 Duplicate detection

```
Scenario 1: Duplicate within the window
Given an earlier transaction exists with merchant M123, order O111, amount 1000
When a second transaction with the same merchant, order and amount arrives within 10 minutes
Then the system shall flag it as Duplicate
And the system shall add it to the Exception Queue

Scenario 2: Same order after the window
Given an earlier transaction exists with merchant M123, order O111, amount 1000
When the same transaction arrives after 10 minutes
Then the system shall not flag it as Duplicate

Scenario 3: Different amount
Given an earlier transaction exists with merchant M123, order O111, amount 1000
When a transaction with order O111 and amount 1500 arrives
Then the system shall not flag it as Duplicate

Scenario 4: Boundary
Given transactions with the same merchant, order and amount at 10:00 and 10:10
When the second transaction is processed
Then the system shall flag it as Duplicate
```

US-02 Exception queue

```
Given a gateway transaction with status SUCCESS and no bank credit by the cut-off time
When the daily reconciliation completes
Then the transaction appears in the Exception Queue as MISSING_SETTLEMENT
And its status is Open
And it is visible to analysts
```

US-03 Late file alert

```
Given the bank file has not arrived by 11:00
When the scheduler checks for files
Then an email alert is sent to the Operations Lead
And the alert names the missing file and the expected time
```

US-06 Resolve with note

```
Given an exception that is In Progress
When the analyst clicks Resolve with an empty note
Then the system shall block the action
And show the message "A resolution note is required"
```

10.4 Definition of Ready

- Story has a clear user, goal and benefit
- Acceptance criteria are written and agreed
- Business rule and examples (JSON) are attached
- Dependencies identified
- Estimated by the team

10.5 Definition of Done

- Code reviewed and merged
- Unit and integration tests pass
- Acceptance criteria verified by QA and BA
- Audit log entry created for each action
- Documentation updated

11. Requirements Traceability Matrix (RTM)

Document ID	RTM-PRE-001
Document title	Requirements Traceability Matrix
Version / Status	1.0 / Baselined
Author / Owner	Business Analyst
Reviewers	Finance Controller, Product Owner, Tech Lead

The RTM links every objective to its business requirement, functional requirements, user story, test cases and UAT scenario. It proves that nothing was built without a need and nothing needed was left untested.

Objective	BR	FR	User story	Test case	UAT	Status
OBJ-1	BR-01	FR-01, FR-02, FR-03	US-02	TC-03, TC-05, TC-12, TC-13	UAT-01, UAT-15	Passed
OBJ-2	BR-02	FR-05, FR-11	US-03	TC-04, TC-07	UAT-02, UAT-11	Passed
OBJ-3	BR-03	FR-04, FR-06	US-01	TC-01, TC-02, TC-06, TC-11	UAT-03, UAT-04, UAT-05, UAT-13	Passed
OBJ-4	BR-04	FR-07, FR-08	US-02, US-06	TC-05, TC-08	UAT-06, UAT-07, UAT-08	Passed
Compliance	BR-05	FR-09	US-05	TC-09	UAT-09, UAT-12	Passed
OBJ-2	BR-06	FR-10	US-04	TC-10	UAT-10	Passed

11.1 Non-functional requirement coverage

NFR	Test case / UAT	Status
NFR-01 Performance	TC-12	Passed
NFR-03 Security	UAT-12	Passed
NFR-04 Audit retention	TC-09 (log is read only); retention verified in configuration review	Passed
NFR-06 Usability	UAT-14	Passed

11.2 Coverage summary

Measure	Value
Business requirements	6 of 6 traced to functional requirements and tests
Functional requirements	11 of 11 traced to at least one test case
Untested requirements	0
Requirements without a business need	0

12. Test Plan, Test Cases and Defect Report

Document ID	TST-PRE-001
Document title	Test Plan, Test Cases and Defect Report
Version / Status	1.0 / Executed
Author / Owner	Business Analyst
Reviewers	QA Lead, Tech Lead, Product Owner
Derived from	FRD-PRE-001

12.1 Test strategy

Level	Purpose	Owner	Environment
Unit testing	Each rule and function works	Developers	Development
Integration testing	File ingestion, engine, database and queue work together	Developers, QA	Test
System testing	End-to-end behaviour against FRD	QA	Test
Performance testing	NFR-01 volume target	QA	Performance
UAT	Business confirms the solution solves the problem	Business users, BA	UAT

12.2 Entry and exit criteria

Entry criteria	Exit criteria
Build deployed to test environment Test data and sample files loaded Test cases reviewed by BA	All Must test cases passed No open critical or high defects Test summary approved by QA Lead and BA

12.3 Test cases

ID	Requirement	Scenario	Input	Expected result	Result
TC-01	FR-04, RULE-01	Duplicate within 10 minutes	Order O100, amount 500, repeated after 5 minutes	Duplicate Flag = Yes	Pass
TC-02	FR-04, RULE-01	Same order after 2 days	Order O100, amount 500, repeated after 2 days	Duplicate Flag = No	Pass
TC-03	FR-03, RULE-02	Clean match	Gateway and bank records agree	Status = MATCHED and auto-closed	Pass
TC-04	FR-05, RULE-03	Missing settlement	Gateway SUCCESS, no bank record	MISSING_SETTLEMENT in queue	Pass
TC-05	FR-07, FR-08	Exception assignment	Analyst assigns an open exception	Owner set; status = In Progress	Pass
TC-06	FR-06, RULE-04	Status mismatch	Gateway SUCCESS, bank REJECTED	STATUS_MISMATCH in queue	Pass
TC-07	FR-11	Late file	Bank file not received by 11 am	Alert sent to Operations Lead	Pass
TC-08	FR-08, RULE-05	Close without note	Resolve with empty note	System blocks closure with message	Pass
TC-09	FR-09	Audit log	Resolve an exception	Log shows user, time, action, old and new status	Pass
TC-10	FR-10	Daily report	Run for a day with known data	Totals agree with source files	Pass
TC-11	FR-04, RULE-01	Boundary: exactly 10 minutes	Same order and amount at 10:00 and 10:10	Duplicate Flag = Yes	Pass (after DEF-007 fix)
TC-12	NFR-01	Performance	1 million records	Completes in under 30 minutes	Pass
TC-13	FR-01, FR-02	Invalid file format	Gateway file with a missing amount column	File rejected; alert sent; no partial load	Pass

12.4 Test execution summary

Measure	Value
Total test cases	13
Passed	13 (one after a fix and retest)
Failed	0 at closure
Defects found	2 (DEF-007 medium, DEF-011 low); both fixed and retested
Open critical or high defects	0

12.5 Defect report (sample)

Defect ID	DEF-007
Title	Duplicate not flagged at exactly 10 minutes
Severity / Priority	Medium / High
Found in	TC-11 (system test)
Steps to reproduce	1. Send a transaction for merchant M123, order O111, amount 1000 at 10:00. 2. Send the same transaction at 10:10. 3. Run reconciliation.
Expected result	Second transaction flagged Duplicate (window is inclusive)
Actual result	Second transaction not flagged
Root cause	Time comparison used less-than instead of less-than-or-equal
Resolution	Developer corrected the comparison; BA confirmed the rule; QA retested
Status	Fixed, retested, closed

13. User Acceptance Testing (UAT) Document

Document ID	UAT-PRE-001
Document title	User Acceptance Testing Document
Version / Status	1.0 / Approved 15-Oct
Author / Owner	Business Analyst
Reviewers	Finance Controller, Product Owner, QA Lead
Derived from	BRD-PRE-001, RTM-PRE-001

13.1 UAT plan

Objective	Confirm that the engine solves the Finance team's real problem and meets business requirements BR-01 to BR-06
Scope	UPI, cards and net banking reconciliation; exception queue; audit log; alerts; reports
Out of scope	Wallet payments, mobile app, merchant portal
Environment	UAT environment, a mirror of production
Test data	Real data from the last 30 days, masked for sensitive fields, plus prepared cases for each rule
Schedule	2 weeks, including a 5-day parallel run of the Excel process and the new engine
Entry criteria	System testing complete; no critical or high defects open; UAT data loaded; testers trained
Exit criteria	All Must scenarios passed; no open critical or high defects; Finance Controller sign-off
Defect handling	Logged in Jira, triaged daily, fixed by Tech team and retested by the BA

13.2 Roles and responsibilities

Role	Who	Responsibility
UAT Lead	Business Analyst	Plans UAT, coordinates testers, logs defects, collects sign-off
Testers	Finance Manager, 2 Operations Analysts, Support representative	Execute scenarios and record results
Approver	Finance Controller	Reviews UAT report and signs off
Technical support	Tech Lead, QA	Fix defects and support the environment

13.3 UAT schedule

Day	Activity
Day 1	Kick-off, tester briefing, environment check
Days 2-5	Execute scenarios UAT-01 to UAT-14; log defects
Days 6-8	Defect fixes and retest
Days 6-10	Parallel run: Excel process and engine side by side (5 working days)
Day 10	UAT summary report
Day 11	Sign-off meeting

13.4 UAT scenarios

ID	Req.	Scenario	Steps	Expected result	Tester	Status
UAT-01	BR-01 FR-03	Daily run matches transactions	1. Load the day's files. 2. Run reconciliation. 3. Compare matched count with the Excel result	Matched count equals Excel	Ops Analyst	Pass
UAT-02	BR-02 FR-05	Same-day visibility of missing settlement	1. Load files with a known missing settlement. 2. Run. 3. Open queue	Item shown as MISSING_SETTLEMENT on the same day	Finance Manager	Pass
UAT-03	BR-03 FR-04	Duplicate within 10 minutes	1. Load two identical transactions 5 minutes apart. 2. Run	Second flagged DUPLICATE	Ops Analyst	Pass
UAT-04	BR-03 FR-04	Same order after 2 days	1. Load identical transactions 2 days apart. 2. Run	Not flagged	Ops Analyst	Pass
UAT-05	BR-03 FR-06	Status mismatch	1. Load SUCCESS at gateway and REJECTED at bank. 2. Run	STATUS_MISMATCH in queue	Ops Analyst	Pass

ID	Req.	Scenario	Steps	Expected result	Tester	Status
UAT-06	BR-04 FR-07	Exception queue content	1. Open queue. 2. Filter by type and date	Only flagged items appear; filters work	Ops Analyst	Pass
UAT-07	BR-04 FR-08	Assign and resolve	1. Assign an exception to self. 2. Enter note. 3. Resolve	Status Resolved; owner and note saved	Ops Analyst	Pass
UAT-08	BR-04 RULE-05	Resolve without note	1. Try to resolve with an empty note	System blocks and shows message	Ops Analyst	Pass
UAT-09	BR-05 FR-09	Audit log	1. Resolve an exception. 2. Open audit log	Entry shows user, time, action, old and new status	Finance Manager	Pass
UAT-10	BR-06 FR-10	Daily summary	1. Open summary. 2. Compare totals with source files	Counts and values agree	Finance Controller	Pass
UAT-11	BR-02 FR-11	Late file alert	1. Withhold the bank file. 2. Wait for the 11 am check	Alert email to Operations Lead	Support rep.	Pass
UAT-12	NFR-03	Access control	1. Log in as Auditor. 2. Try to resolve an exception	Action not available	Finance Manager	Pass
UAT-13	RULE-01	Boundary: exactly 10 minutes	1. Load identical transactions at 10:00 and 10:10	Flagged DUPLICATE	Ops Analyst	Pass
UAT-14	NFR-06	Usability	1. Review one exception and count clicks	3 clicks or fewer	Ops Analyst	Pass
UAT-15	BR-01	Parallel run, 5 days	1. Run engine and Excel each day. 2. Compare results	Same matches; engine finds at least the same mismatches	Ops team	Pass

13.5 UAT defect and enhancement log

ID	Found in	Description	Severity	Status
DEF-007	UAT-13	Duplicate not flagged at exactly 10 minutes	Medium	Fixed and retested
DEF-011	UAT-10	Report total rounded incorrectly in the second decimal	Low	Fixed and retested
ENH-01	UAT-06	Request: filter queue by merchant	Enhancement	Moved to next release (CR-001)
ENH-02	UAT-10	Request: export report to Excel	Enhancement	Moved to next release (CR-001)

13.6 UAT summary report

Measure	Value
Scenarios planned / executed	15 / 15
Passed / failed	15 / 0
Defects found	2 (both fixed and retested)
Open critical or high defects	0
Parallel run result	5 of 5 days matched
Recommendation	Proceed to go-live

13.7 UAT sign-off

Project	Automated Payment Reconciliation Engine
UAT result	Completed
Status	**Approved**
Date	15-Oct
Business comment	"Yes, this solves our problem."
Conditions	None. Enhancements ENH-01 and ENH-02 are planned for the next release

Name	Role	Signature	Date
	Finance Controller		15-Oct
	Product Owner		15-Oct
	Business Analyst		15-Oct

14. Go-Live Checklist, Release Notes and Hypercare Plan

Document ID	PRE-DOC-14
Document title	Go-Live Checklist, Release Notes and Hypercare Plan
Version / Status	1.0 / Completed
Author / Owner	Business Analyst
Reviewers	Finance Controller, Product Owner, Tech Lead

14.1 Go-live checklist

No.	Check	Owner	Status
1	UAT sign-off received	BA	Done
2	Production deployment completed	Tech Lead	Done
3	Gateway and bank file feeds connected	Tech Lead / PSP	Done
4	Reports generated	BA	Verified
5	Files received on schedule	Operations	Verified
6	Matches working against expected volumes	BA / QA	Verified
7	Exceptions created correctly	BA / Analysts	Verified
8	Alerts and access roles tested	Tech Lead / QA	Verified
9	Training and user guide delivered	BA	Done
10	Rollback plan ready	Tech Lead	Ready
11	Hypercare support active (first 2 weeks)	Tech Lead / BA	Active

14.2 Release notes (v1.0)

New in this release

- Automated daily reconciliation for UPI, cards and net banking
- Duplicate, missing settlement and status mismatch detection
- Exception queue with assignment, comments and mandatory resolution note
- Audit log of every automated and manual decision
- Daily reconciliation summary report
- Late and invalid file alerts

Known limitations

- No wallet support
- No merchant filter or Excel export (planned in CR-001)

14.3 Training plan

Session	Audience	Content	Duration
1	Analysts	Exception Queue; resolving with notes	2 hours
2	Operations Lead	Alerts, manual run, file issues	1 hour
3	Finance Controller, Auditor	Reports and audit log	1 hour

14.4 Hypercare and handover

Item	Detail
Hypercare period	2 weeks after go-live
Daily checks	Files received, run completed, exception count, alert emails
Support route	Analyst to Operations Lead to Tech Support
Handover items	User guide, support runbook, known issues list
Benefit tracking	Effort and detection time measured weekly for 3 months

15. Benefits Realization Report

Document ID	PRE-DOC-15
Document title	Benefits Realization Report
Version / Status	1.0 / In progress (3-month tracking)
Author / Owner	Business Analyst
Reviewers	Finance Controller, Product Owner, Tech Lead

15.1 Effort before and after

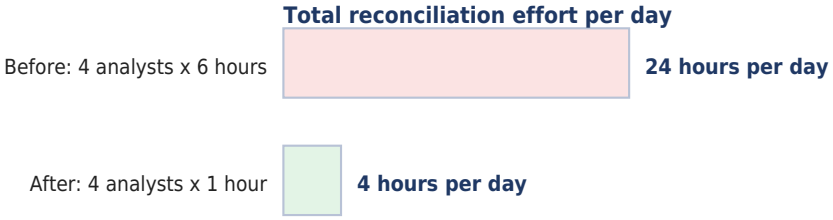


Figure 15.1: Daily reconciliation effort

Measure	Before	After	Change
Analysts	4	4	None
Hours per analyst per day	6	1	-5
Total hours per day	**24**	**4**	**-20**
Effort reduction			**83 percent** (20 divided by 24)
Time to detect a mismatch	About 3 days	Same day (T+0)	About 3 days faster

15.2 Objective scorecard

Objective	Target	Result	Status
OBJ-1 Effort per day	4 hours or less	4 hours	Met
OBJ-2 Detection time	Same day	Same day	Met
OBJ-3 Manual errors	90 percent fewer	Measured over 3 months after go-live	Tracking
OBJ-4 Settlement delay	50 percent fewer delays	Measured over 3 months after go-live	Tracking

15.3 Capacity released

Twenty hours per day are released, about 5,000 hours a year on roughly 250 working days. The capacity moves to exception investigation, merchant queries and analysis. Finance converts hours to rupees using its own loaded cost per hour.

15.4 Lessons learned

- Profile real data in discovery; it exposed the duplicate and status-error patterns early.
- Confirm boundary cases (exactly 10 minutes) in the requirements stage, not in testing; DEF-007 cost a retest cycle.
- A parallel run built trust faster than any presentation.
- Keep stakeholders in the loop with a weekly one-page status; no late surprises at UAT.

16. Supporting Project Documents

Document ID	PRE-DOC-16
Document title	Supporting Project Documents
Version / Status	1.0 / Approved
Author / Owner	Business Analyst
Reviewers	Finance Controller, Product Owner, Tech Lead

16.1 Meeting minutes: Three Amigos session for US-01

Attendees	Business Analyst, Developer, QA
Story	US-01 Duplicate detection
Purpose	Agree rule, examples and tests before development

Topic	Discussion	Decision
Window	Is 10 minutes inclusive?	Yes, inclusive
Amount	Should different amounts count?	No, amount must match
Time zone	Which time zone applies?	IST for all records
Testing	Which cases?	5 minutes, exactly 10 minutes, 2 days, different amount

Action	Owner	Due
Update story with examples and boundary case	BA	Next day
Write test cases	QA	Sprint day 2
Implement rule	Developer	Sprint day 5

16.2 Communication plan

Audience	Message	Channel	Frequency	Owner
Sponsor	Progress, risks, decisions	Status report and call	Weekly	BA / PM
Product Owner	Scope and priorities	Meeting	Twice weekly	BA
Scrum team	Requirement clarity	Stand-up, Jira	Daily	BA
Analysts	Changes to their process	Demo, training	Each sprint	BA
PSP team	File contract changes	Email, call	As needed	BA

16.3 RAID log

Type	ID	Description	Owner	Status
Risk	R1	Gateway file format changes	Tech Lead	Mitigated by schema validation
Assumption	A1	Files arrive by 10 am	Operations Lead	Confirmed with PSP
Issue	I1	Bank file contains blank settlement references	BA	Open; tolerance rule proposed
Dependency	D1	PSP test files before sprint 1	PSP team	Received

16.4 Change request

CR ID	CR-001
Requested by	Finance Manager
Description	Add a merchant filter and Excel export to the exception queue and report
Reason	Faster follow-up with merchants
Impact on scope	Adds two small features
Impact on time and cost	About 4 development days
Risk	Low
Decision	Deferred to the next release
Approved by	Product Owner

16.5 Glossary

Term	Meaning
PSP	Payment Service Provider (the gateway company)
Reconciliation	Proving that two records of the same money agree
Settlement	Transfer of collected funds from the bank to the merchant
MDR	Merchant Discount Rate, the fee charged on transactions
T+0	The same day as the transaction
UAT	User Acceptance Testing
RTM	Requirements Traceability Matrix
MoSCoW	Must, Should, Could, Won't prioritisation
BRD / FRD	Business / Functional Requirements Document

17. Appendix: Health Insurance Claims Auto-Adjudication Mini Pack

Document ID	PRE-DOC-17
Document title	Claims Auto-Adjudication Mini Pack
Version / Status	1.0 / Example
Author / Owner	Business Analyst
Reviewers	Claims Head, Operations Manager

The same BA method applied to a BFSI scenario. Each step mirrors the payment reconciliation project.

17.1 Problem and need

Item	Content
Business problem	Claim processing takes 5 days
Need	Faster claims for simple cases
Stakeholders	Claims Team, Operations Manager, Customers, Compliance
Business requirement	Reduce the turnaround time for simple claims
Functional requirement	The system shall auto-adjudicate claims that meet the simple-claim rules
Out of scope	High-value claims; claims needing medical review
Solution	Rules engine with a manual-review queue

17.2 Business rule

Auto-adjudicate a claim only if: claim amount is below ₹5,000, documents are complete, and fraud score is below the agreed threshold. Otherwise route to manual review.

17.3 Acceptance criteria

```
Given a claim of ₹4,200 with complete documents and fraud score below the threshold
When the claim is submitted
Then the system shall auto-approve it

Given a claim of ₹4,200 with a missing document
When the claim is submitted
Then the system shall route it to manual review

Given a claim of ₹6,000
When the claim is submitted
Then the system shall route it to manual review
```

17.4 Result

Measure	Before	After
Processing time for simple claims	5 days	1 day
Improvement		80 percent faster